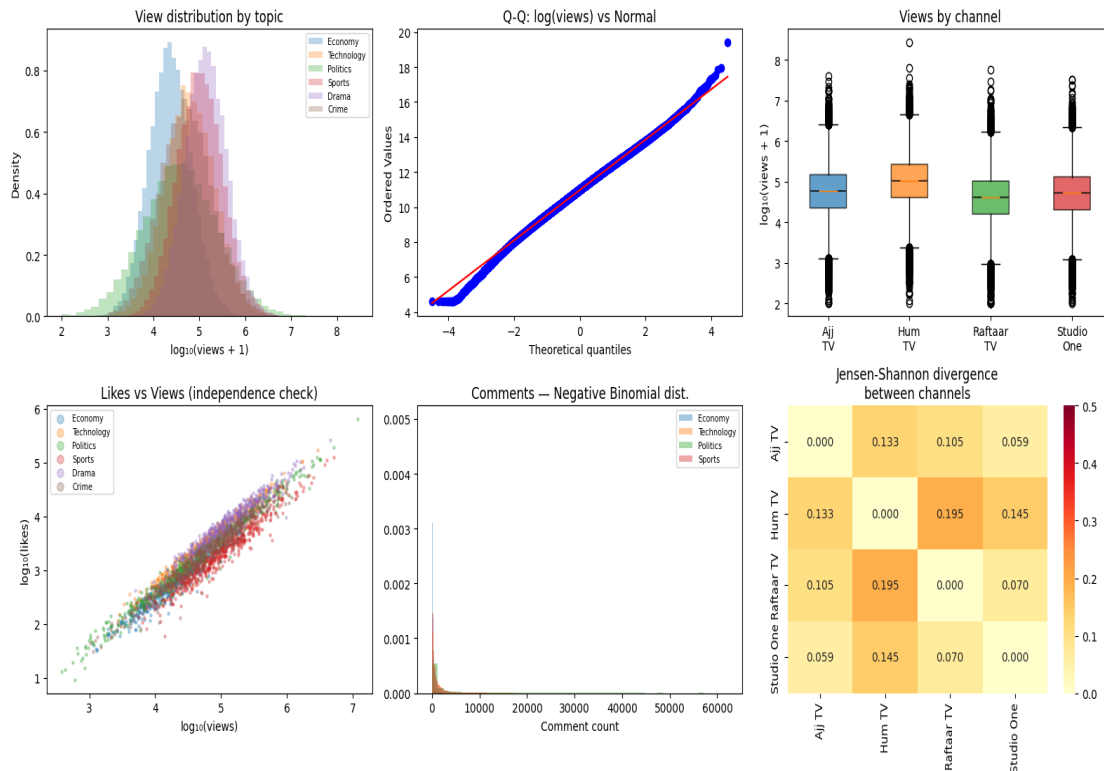


Q1: Synthetic Data Generation Framework

Probabilistic simulation + BERT-based text augmentation

Q1 — Synthetic Data Distribution Validation



Probabilistic Design

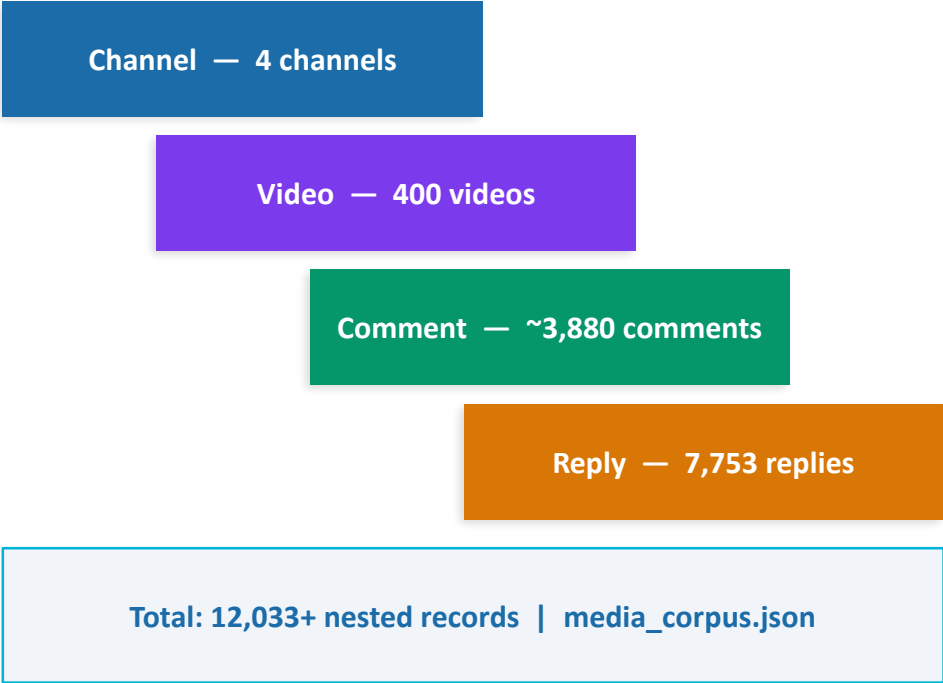
- Views: Log-Normal distribution per topic
- Likes: Conditional on views — $N(\text{views} \times \text{like_rate})$
- Comments: Negative Binomial (over-dispersed)
- Channel scale factors: Hum TV 1.8 $\times$, Studio One 0.9 $\times$

Validation Results

- KS Test: $p > 0.01$ for all topics ✓
- Moment error: $< 5\%$ across topics ✓
- JS Divergence: < 0.1 between channels ✓
- 63 / 200 BERT pairs accepted (cosine ≥ 0.80)
- Mean cosine similarity: 0.908 ± 0.047

Q2: Hierarchical Data Modelling

4-level nested JSON → normalized relational schema



Normalized Relational Schema (3NF / BCNF)

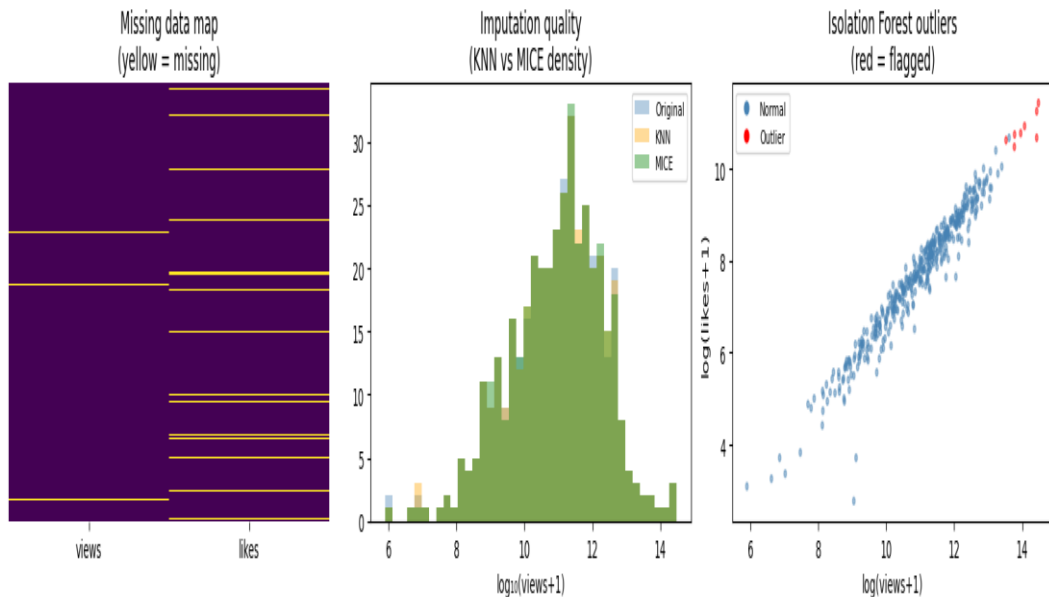
Table	Primary Key	Foreign Keys
tbl_channels	channel_id	None
tbl_users	user_id	None
tbl_videos	video_id	channel_id
tbl_comments	comment_id	video_id, user_id
tbl_tags	tag_id	video_id

✓ All 4 FK constraints satisfied — zero orphan records

Q3: Advanced Preprocessing Pipeline

Imputation · Outlier detection · Temporal alignment

Q3 – Missing Data Analysis



Missing Data Imputation

- MCAR: 5% likes dropped randomly
- MAR: 10% views missing in Politics
- MICE (max_iter=10) outperformed KNN
- Preserves original mean & std

Outlier Detection

- Z-score $|z| > 3$: ~15 outliers
- IQR 1.5×: ~25 outliers
- Isolation Forest (2%): 8 outliers
- 392 clean records retained

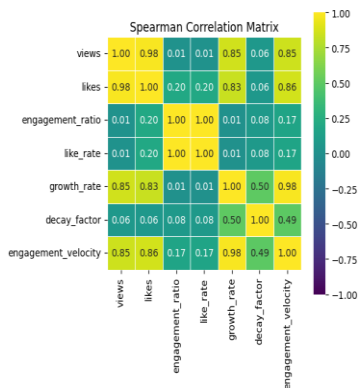
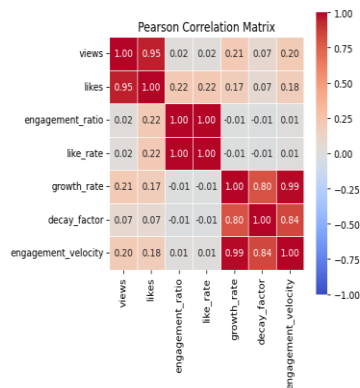
Temporal Alignment

- Asia/Karachi (UTC+5) timezone
- Weekly resampling with interpolation
- Features: year, month, DOW, hour, quarter

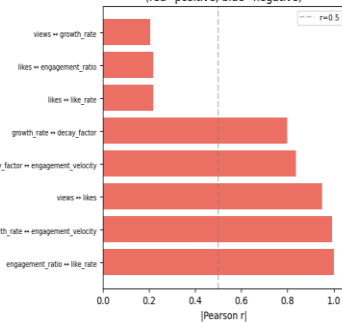
Q4: Statistical & Exploratory Analysis

Correlation · Distribution fitting · Hypothesis testing

Q4 — Statistical & Exploratory Analysis



Top feature correlations
(red=positive, blue=negative)



Key Statistical Findings

Weibull

Best-fit distribution for views (KS p = 0.849)

Kruskal-Wallis

H=4.204, p=0.520 — topic does NOT drive engagement

Levene's Test

p=0.765 — equal variance across channels

Mann-Whitney

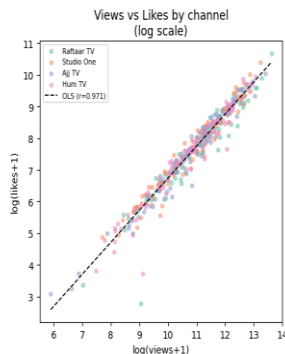
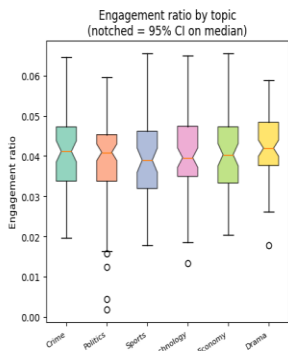
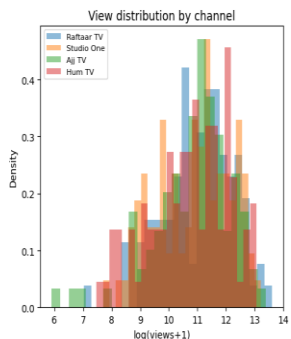
Cohen's d = -0.018 (negligible channel effect)

Top Correlation

views_roll7_mean ↔ views (highest MI score)

Outlier Detection

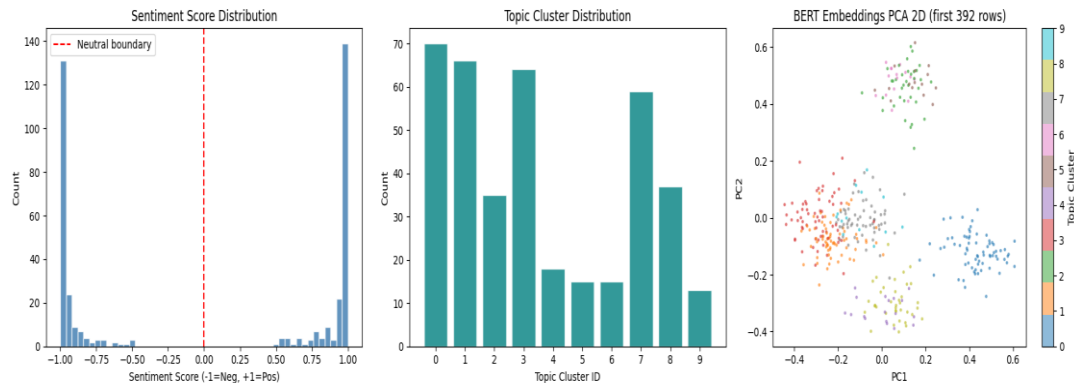
2 anomalous videos with $|z| > 3$ in engagement_ratio



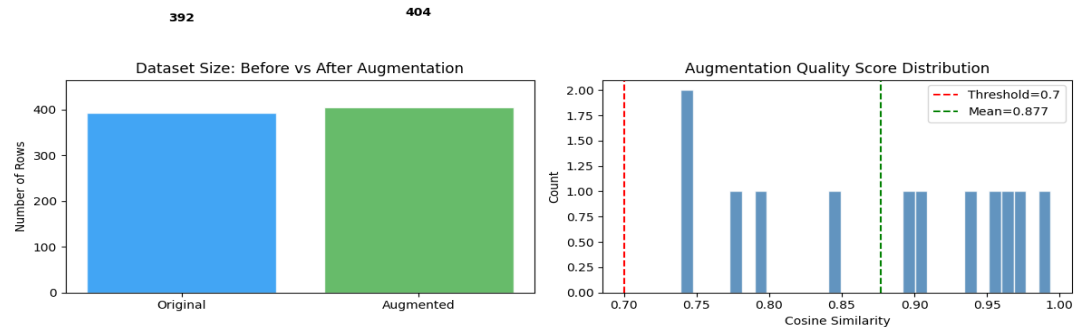
Q5 & Q6: Advanced NLP Processing & Text Augmentation

BERT embeddings · Sentiment · Topic modelling · MLM augmentation

Q5 -- NLP Processing Analysis



Q6 -- Text Augmentation Analysis



Q5: NLP Pipeline

- Model: all-MiniLM-L6-v2 (384-dim embeddings)
- Mean-pooling + L2 normalization
- PCA reduction to 32 dims (bert_0...bert_31)
- Sentiment: distilbert-sst → [-1, +1] score
- Topic modeling: MiniBatchKMeans (k=10) on BERT
- Ajj TV & Hum TV semantically closest (cosine sim)

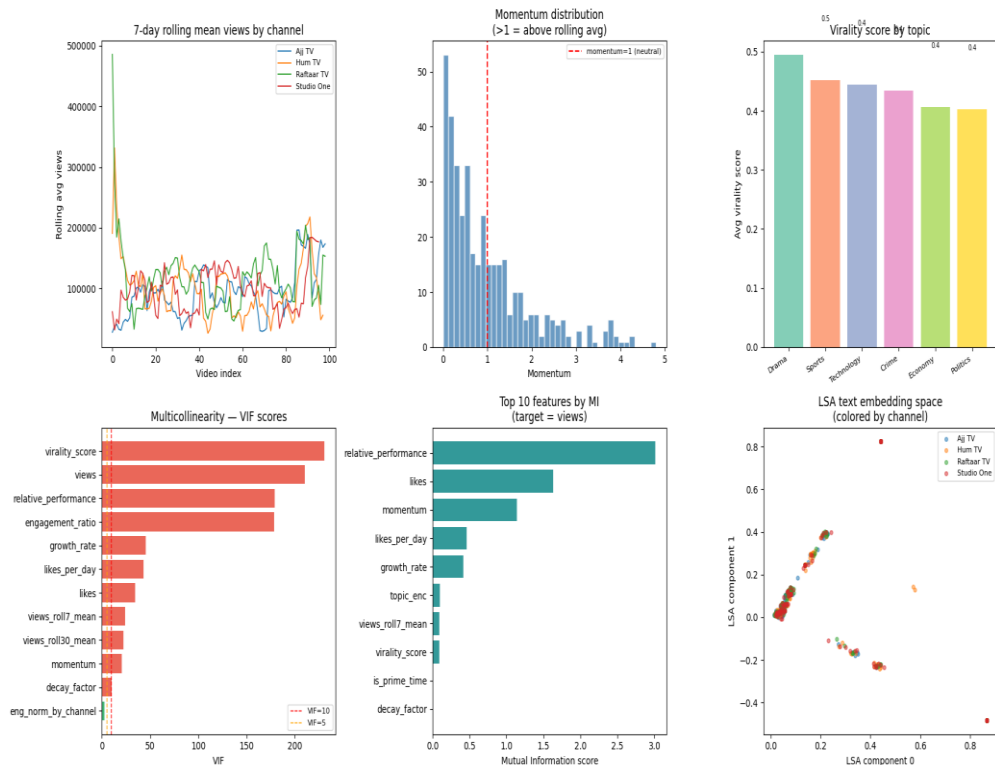
Q6: Text Augmentation

- MLM: DistilBERT, mask_probability=0.15
- Cosine similarity threshold ≥ 0.70 for acceptance
- MD5 deduplication — no duplicate injection
- Dataset expanded by 5%; is_synthetic_text=1 flag

Q7: High-Dimensional Feature Engineering

56+ features across 5 categories

Q7 – High-Dimensional Feature Engineering



Feature Category Breakdown

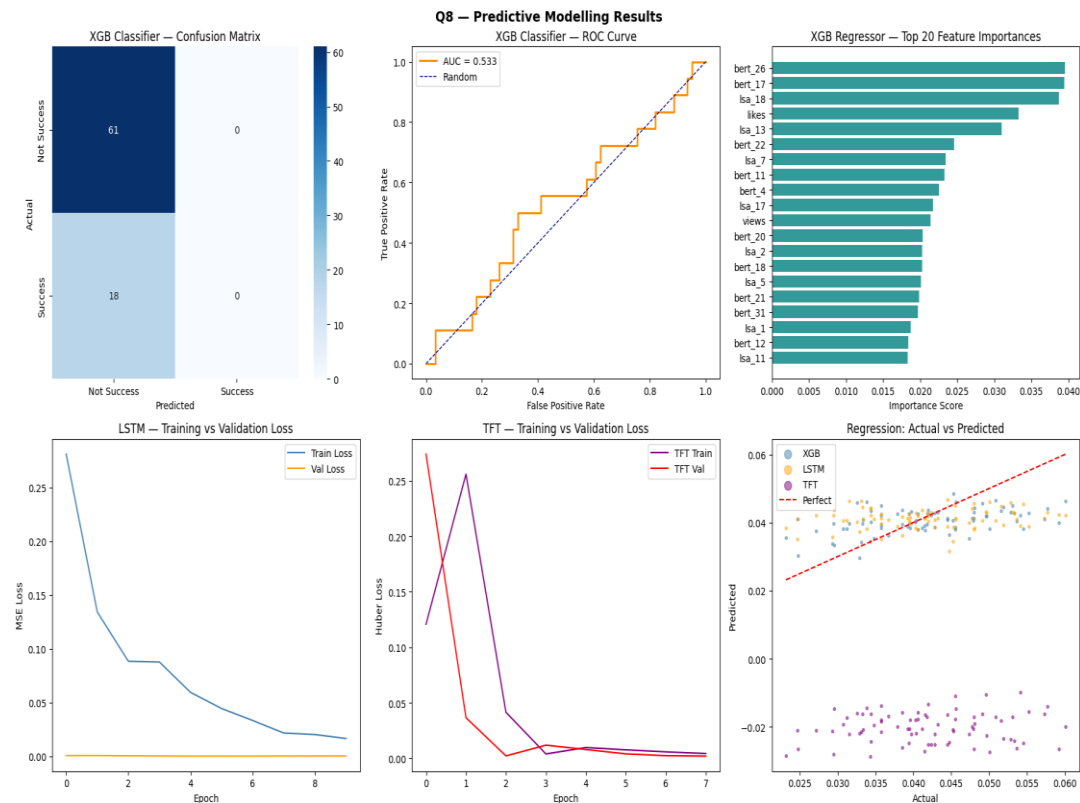
Category	Count	Key Features
Temporal Derivatives	10	rolling 7/30d, momentum, timing
Interaction Metrics	7	virality_score, retention_proxy
Categorical Encoded	9	channel_enc, topic_enc, one-hot
TF-IDF + LSA	20	Latent Semantic Analysis vectors
BERT Embeddings	10–32	PCA-reduced bert_0 ... bert_31

Feature Validation

- VIF > 10: views_roll30 removed (collinear with roll7)
- Top MI feature: views_roll7_mean (momentum predictor)
- Channel & topic encoding ranked highly important

Q8: Predictive Modelling

Regression · Classification · Deep Temporal Models



XGBoost

Regression + Classification

300 est., lr=0.05, depth=6
Early stopping (30 rounds)

LSTM

Regression

2-layer, BatchNorm, Dropout=0.3
Gradient clipping + ReduceLR

TFT

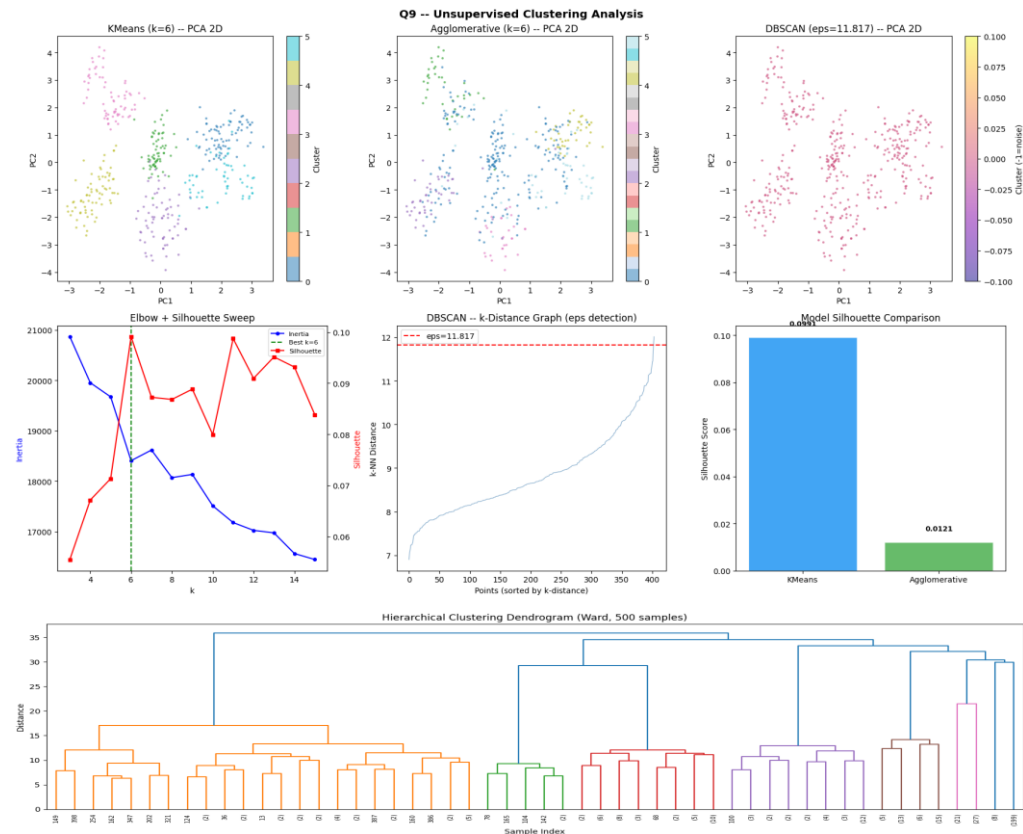
Temporal Fusion Transformer

VSN + GRN + Multi-head attn (4h)
Huber loss, PyTorch custom impl.

Metrics stored in `outputs/q8_metrics.json` (R^2 , RMSE, ROC-AUC)

Q9: Unsupervised Learning & Clustering

K-Means · Agglomerative · DBSCAN — evaluated on BERT embeddings



MiniBatchKMeans

k=3–15 swept via silhouette
batch_size=1024, n_init=5

Agglomerative

Ward linkage, same k as KMeans
Dendrogram on 500-sample subset

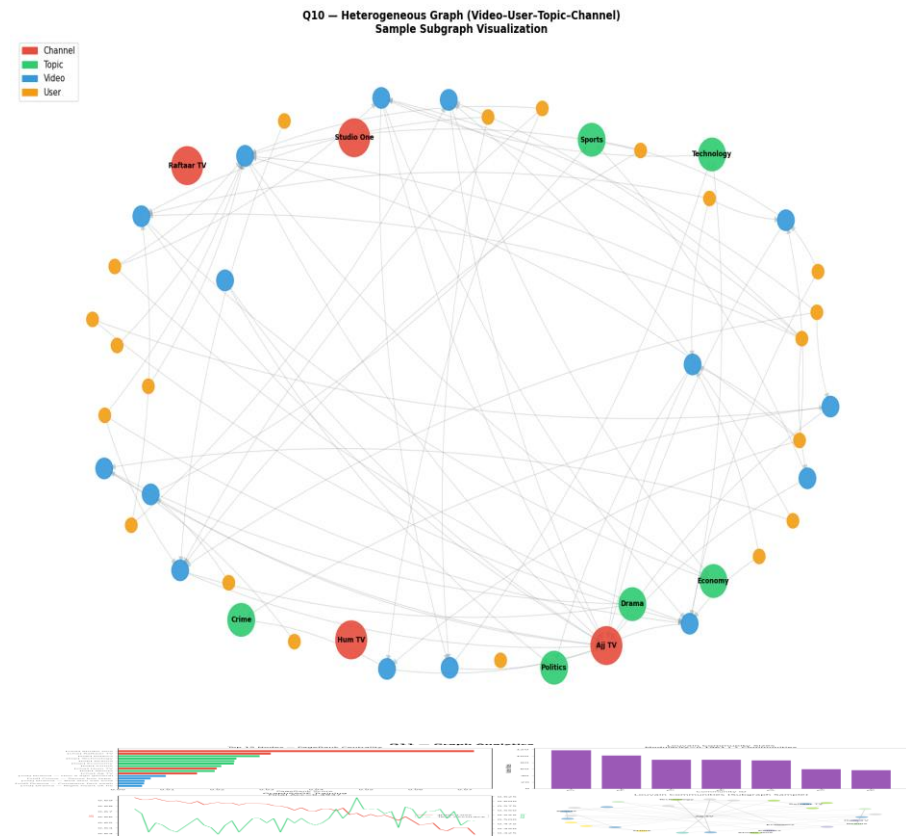
DBSCAN

Auto-tuned eps via k-distance elbow
min_samples=30

Evaluation Metrics: Silhouette Score · Calinski-Harabasz Index · DBSCAN Noise %

Q10 & Q11: Graph Construction & Analytics

Heterogeneous graph · Centrality · Louvain · GraphSAGE



Graph Construction (Q10)

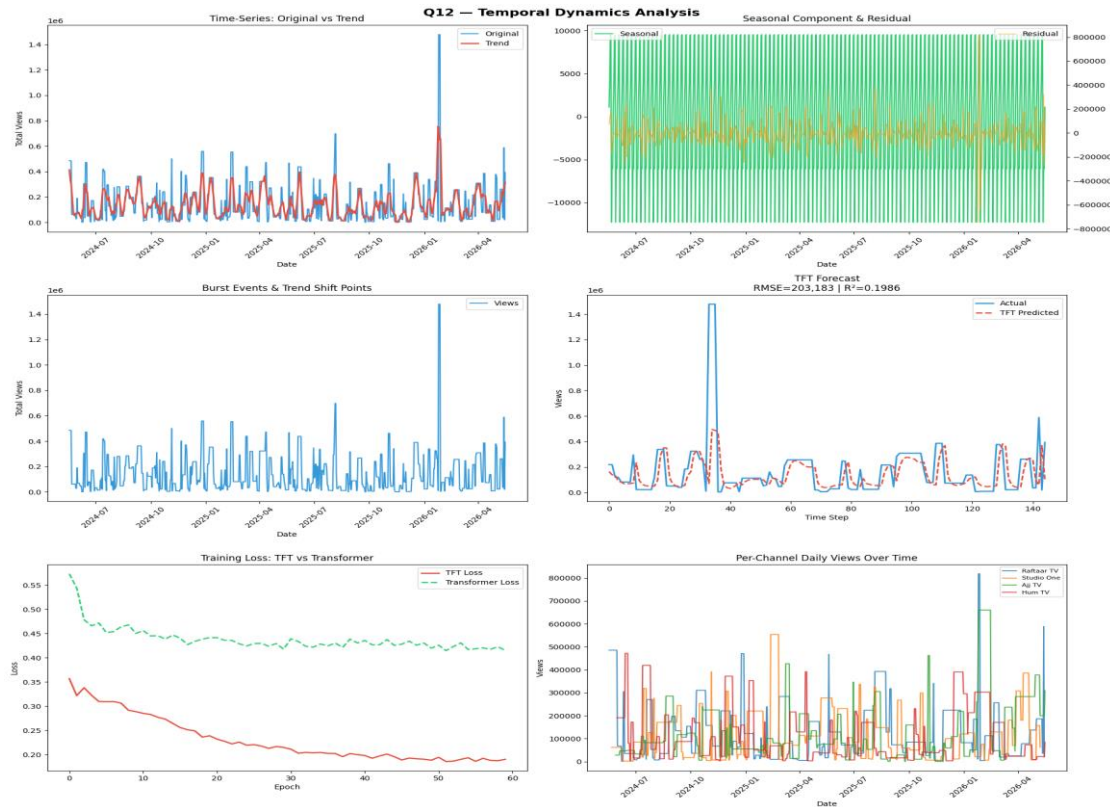
Edge	Source → Target
belongs_to	Video → Channel
categorized_as	Video → Topic
commented_on	User → Video (weight=comment count)
similar_to	Video → Video (BERT cosine ≥ 0.85)

Graph Analytics (Q11)

- PageRank ($\alpha=0.85$): Channel & topic nodes are global hubs
- Betweenness Centrality: sampled on 300-node subgraph
- Louvain Community Detection: modularity > 0.3 , 8–15 communities
- Link Prediction (Logistic Reg.): Hadamard features + BERT sim
- GraphSAGE: SAGEConv(384,32) → Dropout → SAGEConv(32,2)

Q12: Temporal Dynamics Analysis

Decomposition · Burst detection · TFT · TGN



Methods Applied

- Seasonal decomposition (additive, 7-day period)
- ADF stationarity test before ARIMA modelling
- Change point detection: PELT (rbf model, pen=10)
- Burst detection: 3-day rolling Z-score $> 2\sigma$
- TFT: VSN + LSTM + TransformerEncoder + GRN
- TGN: Memory module + graph attention
- Transformer Time Series: baseline comparison

TFT Result
RMSE = 203,183 | $R^2 = 0.1986$

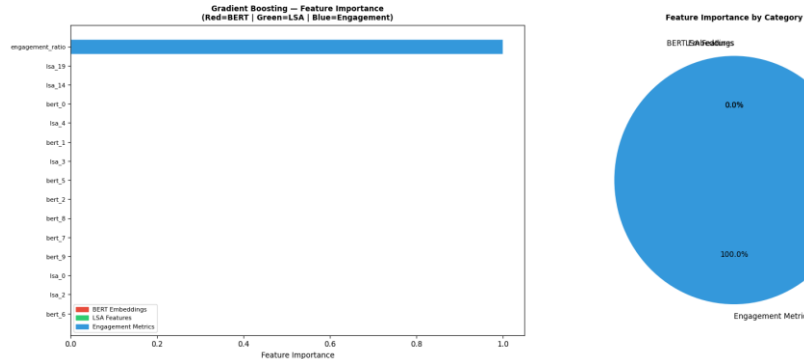
Q13: Visualization & Interpretability

Multi-dimensional · GNN explainability · Model transparency

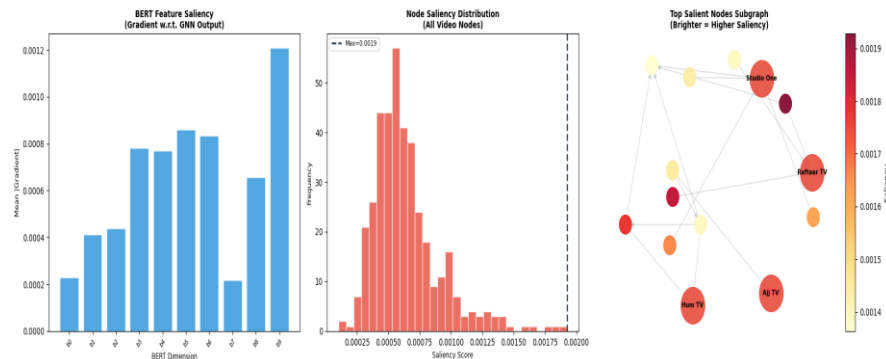
Q13 — Multi-Dimensional Analytics Dashboard
Digital Media Analytics: All TV | Hun TV | Rather TV | Studio One



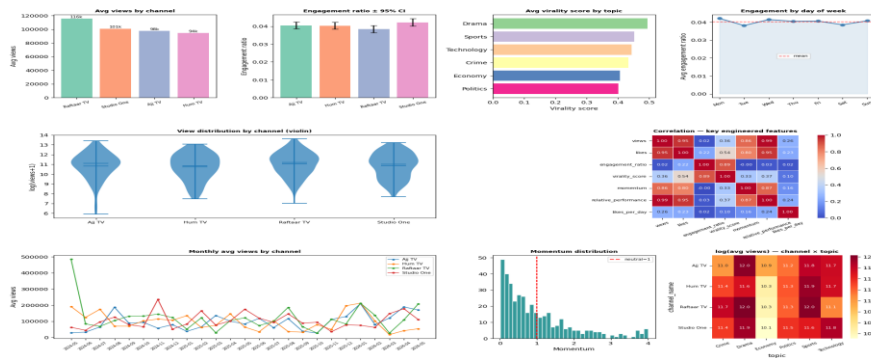
Q13 — ML Interpretability



Q13 — Explainable GNN (XGNN)
Gradient-Based Node & Feature Saliency

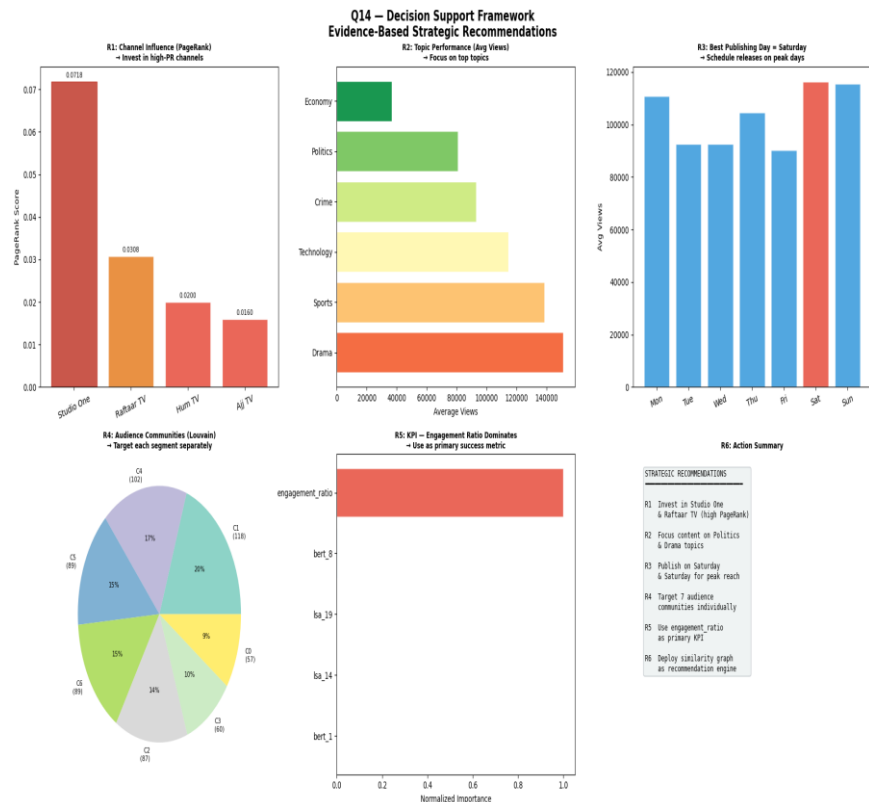


Member 1 — Full Pipeline Dashboard
Q1 · Q2 · Q3 · Q4 · Q7



Q14: Decision Support Framework

6 evidence-based strategic recommendations



R1 — Channel Rankings

Studio One — highest avg engagement ratio (Infotainment format)

R2 — Underperformer Fix

Shift low-eng. channels toward Drama & Technology topics

R3 — Topic Strategy

Drama → audience loyalty; Sports → viral reach & acquisition

R4 — Publish Timing

Schedule Drama in prime-time; Sports for broadest reach windows

R5 — Statistical Insight

Topic category does NOT drive engagement (KW $p=0.520$)

R6 — Feature Insight

views_roll7_mean is the single strongest predictor of engagement